CASE STUDY 1

TARGET: An insurance company wants to decide how to invest its assets for the next five years.

CHALLENGE: What if the future is different from what we expect?

SOLUTION: Analyzing three different scenarios.

1: BASELINE SCENARIO	2: JAPANESE SCENARIO	3: HIGH INFLATION SCENARIO
• The economy grows normally. • Interest rates behave as expected. • Inflation remains moderate. • Everything is relatively stable.	This refers to what happened in Japan after the 1990s. • Very low interest rates • Slow economic growth • Deflation (falling prices) • Long economic stagnation -**Dangerous** because they earn very little on their investments while still having to meet long-term guarantees	• Very high inflation. • Interest rates rise sharply. • Markets become volatile. • Bond prices fall. • Some investments lose value.
Implementing a *strategy* based on 3 scenarios		
Optimize for the baseline	**Optimize only for Japan**	**Balance all three scenarios**
Optimizing only for the expected scenario maximizes returns but may result in insolvency under adverse economic conditions. (Japan and high inflation scenario)	One possible strategy is to invest primarily in government bonds, which tend to perform relatively well during prolonged periods of low interest rates and economic stagnation. However, this strategy performs poorly when inflation rises sharply because fixed coupon payments lose purchasing power and bond prices typically decline as interest rates increase. Therefore, protecting against one specific risk may increase exposure to another.	Here, the insurer adopts a more balanced strategy by reducing exposure to risky assets (de-risking) and incorporating inflation-hedging investments. The result is a portfolio that performs robustly under each scenario.

Final Improvement

The insurer strengthens its strategy even more by adding:

- **External reinsurance** (transferring part of the risk to another insurer).
- **Intra-group reinsurance/retrocession** (sharing risk within its corporate group).

Main lesson

The objective is not maximum expected profit, but robust solvency across a range of future scenarios.

Actuarial insight

Actuarial risk management is about building robust strategies—not maximizing returns in a single scenario but maintaining resilience and solvency across multiple plausible futures.

International Actuarial Association. (2025). *Stress testing and scenario analysis*. https://actuaries.org/app/uploads/2025/04/StressTestingPaper.pdf

CASE STUDY 2

TARGET: Build a pandemic scenario for stress testing to assess how an insurance company or bank would perform during a pandemic.

CHALLENGE: Design a plausible scenario that meaningfully tests resilience without being unrealistically optimistic or catastrophic.

SOLUTION: Define the scenario, estimate its financial impacts, and evaluate the institution's resilience, solvency, and risk exposure.

EXAMPLE

SCENARIO 1: MILD INFLUENZA

The pandemic begins in Asia and rapidly spreads worldwide, causing approximately five million deaths. Central banks respond quickly by lowering interest rates and injecting liquidity into the financial system. Although financial markets experience a sharp decline, they gradually recover, making the pandemic a severe but ultimately manageable crisis.

SCENARIO 2: SEVERE INFLUENZA

Unlike the previous scenario, this pandemic unfolds in two waves. The first wave is relatively moderate, and the vaccine provides only partial protection. However, the virus later mutates, triggering a far more lethal second wave that results in approximately 44 million deaths, widespread panic, and near-collapse of financial markets. In addition to increased mortality, insurers also face significant liability risk due to legal claims related to vaccine side effects.

SCENARIO 3: NO INFLUENZA

This scenario involves a completely new virus with a very low infection rate—only about 1% of the population is infected—but an extremely high fatality rate of 80%. Despite the limited spread, fear triggers severe economic disruption: international trade slows dramatically, flights are cancelled, and financial markets nearly shut down. This illustrates that economic impact depends not only on infections or deaths, but also on how people, governments, and markets respond to the crisis.

DEFINING THE SCENARIO:

- What type of pandemic? Influence, COVID, resistant bacteria, …
- What is their intensity? 2 million deeds, 10 million, 60 million, …
- How long does it last? 6 months, 3 years
- How would actuarial variables affect? Mortality, Morbidity, Disability, Frequency of accidents, Severity of claims…
- How would affect the markets? Because it triggers Stock market crash, Unemployment, Lower GDP, Lower interest rates, Wider spreads, Currency depreciation, …

CALCULATING FINANCIAL IMPACT

How would affect my company?

- More deeds so more claims, more hospitality so more medical expenses…
- What will happen with our investments?

Key point: Your assumptions define the severity of the stress test. For example, you may assume mortality rates of 0.05%, 2%, or 6% to evaluate different scenarios. However, scenarios should be plausible, neither overly optimistic nor excessively pessimistic.

Main Lesson

A scenario is not just an event, it is a coherent narrative that connects an initial shock to its financial, operational, and strategic impacts, helping organizations evaluate resilience and prepare for uncertainty before a crisis unfolds.

Actuarial Insight

A pandemic should not be viewed solely as a health event, but as a systemic risk affecting multiple dimensions simultaneously. It impacts biometric risks (mortality, morbidity, and disability), financial risks (equities, bonds, interest rates, exchange rates, and credit spreads), operational risks (business disruptions and supply chains), legal and reputational risks, as well as liquidity and solvency through increased claims and capital pressures.

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